

Non-Bank Financial Intermediation and Systemic Risk

Research Project Outlines

Project 1 The Shadow Leverage Map

Repo & Derivatives Network Analysis

Project 2 Mapping the Channels

NBFI Amplification of Monetary Policy

Project 3 Contagion Across Borders

NBFI Stress & the Global Financial Cycle

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February 2026

Table of Contents

1. Executive Summary

2. Project 1 --The Shadow Leverage Map

- 2.1 Research Questions
- 2.2 Data Requirements
- 2.3 Empirical Framework
- 2.4 Expected Contributions

3. Project 2 --Mapping the Channels

- 3.1 Research Questions
- 3.2 Data Requirements
- 3.3 Empirical Framework
- 3.4 Expected Contributions

4. Project 3 --Contagion Across Borders

- 4.1 Research Questions
- 4.2 Data Requirements
- 4.3 Empirical Framework
- 4.4 Expected Contributions

5. Shared Econometric Toolkit

6. Implementation Status & Codebase

7. References

1. Executive Summary

This document outlines three complementary research projects investigating the role of non-bank financial intermediaries (NBFIs) in systemic risk and financial stability. Together, they form a unified research programme that addresses key gaps identified by the BIS, FSB, and academic literature on the growing role of shadow banking in the global financial system.

The projects share a common methodological foundation --combining network analysis, time-varying econometrics (DCC-GARCH), and cutting-edge tail-risk tools (quantile VARs, location-scale models, Growth-at-Risk) --but each targets a distinct policy question:

1. Project 1 (Shadow Leverage Map) asks: How much hidden leverage exists in the NBFI sector, and can we detect it from publicly available repo and derivatives data?
2. Project 2 (Mapping the Channels) asks: How do NBFIs amplify monetary policy transmission, and why was the real economy resilient despite amplified financial tightening in 2022-23? This directly extends BIS Bulletin 116 (Banerjee, Hofmann, Ng & Pinter, 2025).
3. Project 3 (Contagion Across Borders) asks: How does NBFI stress in one jurisdiction spill over to others via dollar funding channels, portfolio flows, and the global financial cycle?

Each project is implemented as a self-contained Jupyter notebook with full synthetic data, econometric estimation, and publication-quality visualizations, supported by a shared Python codebase. The synthetic data mirrors the structure of real data sources (OFR, DTCC, ECB, BIS IBS, FSB) to facilitate seamless transition to live data.

2. Project 1 --The Shadow Leverage Map

Tracing Hidden Non-Bank Exposures Through Repo and Derivatives Data

2.1 Research Questions

This project addresses three interconnected questions:

Primary Research Question

RQ1: Can we construct a reliable estimate of the hidden leverage embedded in NBFi balance sheets using publicly available market data (repo volumes, derivatives notionals, counterparty concentration metrics)?

Secondary Questions

- RQ2: What is the network topology of bank-NBFi exposures through repo and derivatives markets, and how has it evolved since the GFC?
- RQ3: Do clusters of interconnected NBFis create systemic exposure that individual balance sheets fail to reveal? Can we detect these 'hidden leverage clusters' from eigenvalue analysis of the exposure network?
- RQ4: Does shadow leverage predict worse tail outcomes (Growth-at-Risk)? Is the relationship nonlinear --i.e., does it matter more during stress?

2.2 Data Requirements

Source	Variables	Frequency	Coverage
OFR US Repo	Bilateral volumes, rates, haircuts by counterparty type	Daily	US tri-party & bilateral, 2013-present
DTCC Derivatives	CDS/IRS notional, counterparty concentration	Weekly	US OTC derivatives, 2010-present
ECB MMSR	Secured/unsecured money market volumes & rates	Daily	Euro area, 2016-present
FSB NBFi	Sector-level assets by jurisdiction	Annual	29 jurisdictions, 2002-present
BIS Derivatives	OTC notional outstanding by counterparty sector	Semi-ann.	Global, 1998-present

For the prototype implementation, we generate synthetic data that mirrors the structure, temporal patterns, and cross-sectional variation of these real sources, including crisis episodes (March 2020, September 2022). All synthetic data uses fixed random seeds for reproducibility.

2.3 Empirical Framework

Step 1: Implied Leverage Estimation

We estimate implied leverage for each NBFi type from observable market data. For an entity i with estimated equity E_i , observed repo borrowing R_i , and derivative notional exposure D_i :

$$\text{Implied_Leverage}_i = (E_i + R_i + \text{delta} * D_i) / E_i$$

where delta is a notional-to-exposure conversion factor calibrated by derivative type (CDS vs. IRS vs. FX). This extends the BIS approach of using gross notional as a leverage proxy (BIS 2025, Chapter II).

Step 2: Network Construction

We build quarterly bipartite bank-NBFI networks from counterparty exposure data. Edge weights represent bilateral exposure amounts. We compute: degree/strength centrality, eigenvector centrality, betweenness, PageRank, and one-mode projections (Abad et al. 2022).

Step 3: Hidden Leverage Detection

We use eigenvalue analysis of the exposure-weighted adjacency matrix. The largest eigenvalue of the bank-NBFI network tracks aggregate systemic exposure: when it rises, the system is becoming more interconnected and amplification risk increases. We also identify clusters of NBFIs whose combined leverage creates systemic risk invisible at the individual level.

Step 4: Tail-Risk Econometrics

- Quantile connectedness (Ando et al. 2022): Compare network connectedness at $\tau=0.05$ (left tail / stress) vs $\tau=0.50$ (median). Higher tail connectedness = more contagion in crises.
- Growth-at-Risk (Adrian et al. 2019): Quantile regression of future economic outcomes on the shadow leverage index. Tests whether hidden leverage predicts worse tail outcomes.
- Tail risk amplification: Does the interaction of stress x shadow_leverage worsen the 5th percentile of outcomes? Estimated across the full quantile spectrum.
- Variance ratio test: Do countries/periods with higher shadow leverage exhibit significantly more volatile outcomes?

2.4 Expected Contributions

1. First working prototype for estimating NBFI leverage from publicly available data --addresses the key data gap identified by the FSB and BIS.
2. A 'Shadow Leverage Index' that can serve as an early warning indicator for systemic build-up in the non-bank sector.
3. Evidence on whether hidden leverage is a tail-risk amplifier --directly relevant for macroprudential policy calibration.

3. Project 2 --Mapping the Channels

Network Analysis of Non-Bank Amplification in Monetary Policy Transmission
Building on BIS Bulletin 116 (Banerjee, Hofmann, Ng & Pinter, December 2025)

3.1 Research Questions

Primary Research Question

RQ1: Banerjee et al. (2025) show that OFIs amplify monetary policy transmission to long-term yields, financial conditions, and GDP --but with wide confidence bands and at an aggregate sector level. Can we decompose this effect: WHICH types of OFIs amplify most, THROUGH WHICH channels, and UNDER WHAT conditions?

Secondary Questions

- RQ2: Is amplification nonlinear? Does it intensify in the tails of the distribution (quantile VAR) or only under specific states (high leverage, low liquidity)?
- RQ3 (The Puzzle): Why was the real economy resilient during the 2022-23 hiking cycle despite financial conditions tightening more than in previous cycles? Can network topology explain the containment of amplification within the financial sector?
- RQ4: Does the NBFI sector affect the conditional variance (not just the mean) of economic outcomes? If so, NBFIs create fatter tails even when mean transmission appears moderate.

Positioning Relative to Banerjee et al. (2025)

We explicitly build on --rather than compete with --BIS Bulletin 116. Their paper establishes the aggregate fact (OFIs amplify, ICPFs dampen). We provide the micro-level decomposition that their framework cannot offer. Our value-add:

- Their approach: sector-level cross-country panels with linear local projections.
- Our approach: entity-level network analysis + quantile methods for nonlinearity.
- Their gap: 'considerable uncertainty about the extent of this effect remains.'
- Our contribution: decompose that uncertainty into identifiable channels.

3.2 Data Requirements

Source	Variables	Use
Central bank rates	Policy rates (Fed, ECB, BoE, BoJ), forward guidance dates	Monetary policy shock identification
Financial returns	Daily returns: banks, HFIs, bond funds, MMFs, insurance, p	DCC-GARCH, QVAR, connectedness
Financial conditions	FCI (Goldman, Chicago Fed), VIX, credit spreads, term pre	State variables, GaR regressions
Balance sheet data	NBFI leverage ratios, AUM, repo borrowing by sector	VaR-constraint model calibration
Real economy	GDP growth, credit growth, investment, unemployment (Q)	Outcome variables for transmission
FSB NBFI Monitor	Sector assets by jurisdiction	Cross-country NBFI penetration

3.3 Empirical Framework

The framework has five integrated modules, each targeting a link in the transmission chain:

Module 1: VaR-Constraint Channel (Adrian & Shin 2010)

Estimate procyclicality of leverage by sector. Run fire-sale counterfactuals comparing bank-only vs. bank+NBFI systems. Key prediction: hedge funds exhibit the highest procyclicality ($\beta > 1$), followed by finance companies.

Module 2: DCC-GARCH Dynamic Correlations (Engle 2002)

Estimate time-varying correlations between bank and NBFI sector returns. Test for correlation breakdown during the hiking cycle (Forbes & Rigobon 2002 style). Track bank-HF, bank-MMF, bank-insurance correlation paths.

Module 3: Quantile VAR [Key Innovation]

Estimate a QVAR system [policy_rate, HF_returns, bond_fund_returns, bank_returns, FCI, GDP_growth] at $\tau = \{0.05, 0.25, 0.50, 0.75, 0.95\}$. Compute quantile impulse response functions (Montes-Rojas 2019) and quantile connectedness (Ando, Greenwood-Nimmo & Shin 2022).

Key hypothesis: At the 5th percentile (stress), the IRF of HF returns to a policy rate shock is 2-3x larger than at the median. This is the nonlinear amplification that Banerjee et al.'s linear framework cannot capture.

Module 4: Location-Scale Growth-at-Risk

Model both the conditional mean AND conditional variance of future GDP growth as functions of financial conditions and NBFI variables. If NBFI leverage increases the conditional variance, NBFIs create fatter tails even when mean transmission appears moderate.

Module 5: Solving the Banerjee et al. Puzzle

Use sector-level connectedness decomposition to show that amplification was contained within OFI clusters. The Diebold-Yilmaz network reveals high within-OFI connectedness but low OFI-to-real-economy spillovers, explaining the resilience puzzle.

3.4 Expected Contributions

1. Decomposition of the aggregate OFI amplification documented by Banerjee et al. (2025) into entity-level and channel-level components.
2. First application of quantile connectedness (Ando et al. 2022) to monetary policy transmission --capturing nonlinear/tail dynamics.
3. Resolution of the 2022-23 resilience puzzle through network topology analysis.
4. Evidence that NBFIs affect the conditional variance of outcomes (location-scale), not just the conditional mean --a distinction with direct macroprudential implications.

4. Project 3 --Contagion Across Borders

How NBFI Stress in One Jurisdiction Spills Over via the Global Financial Cycle

4.1 Research Questions

Primary Research Question

RQ1: Through which channels does NBFI stress in one jurisdiction transmit to financial conditions in other countries? Specifically, how do (a) dollar funding, (b) portfolio rebalancing, and (c) the global financial cycle interact to create cross-border contagion?

Secondary Questions

- RQ2: Is cross-border contagion asymmetric --stronger during stress episodes than during calm periods? (Quantile connectedness across countries.)
- RQ3: Does NBFI penetration in a country increase its vulnerability to external shocks transmitted through the global financial cycle (Miranda-Agrippino & Rey 2020)?
- RQ4: How does a major NBFI failure (e.g., Archegos-style event) cascade across borders? What is the role of prime brokerage concentration?
- RQ5: Do NBFI-related capital flows increase the conditional variance (fat tails) of emerging market financial variables?

4.2 Data Requirements

Source	Variables	Use
BIS IBS	Cross-border claims by country x sector (bank/NBFI)	Exposure network construction
BIS Derivatives	OTC notional by counterparty country and sector	Derivative exposure mapping
Portfolio flows	Gross in/outflows by investor type (EPFR, IMF BoP)	Portfolio flow channel analysis
FX swap basis	USD vs EUR, GBP, JPY, CHF cross-currency basis	Dollar funding stress indicator
Equity indices	MSCI country indices (10+ countries)	GFC extraction, return spillovers
Bond yields	10Y government bond yields across countries	Yield spillover analysis
Fed swap lines	Swap line usage by central bank counterparty	Dollar liquidity provision

4.3 Empirical Framework

The framework is organized around the "Triple Helix" of cross-border NBFI contagion:

Channel 1: Dollar Funding

NBFIs borrow in USD via repo and FX swaps. When funding tightens (wider FX swap basis), they deleverage, creating fire-sale pressure. We estimate quantile regressions of local financial conditions on USD funding stress, interacted with the country's NBFI exposure. The relationship is expected to be nonlinear: much stronger in the left tail.

Channel 2: Portfolio Flows

Global investment funds rebalance portfolios across borders. NBFI flows are more volatile and procyclical than bank flows. We compute Diebold-Yilmaz connectedness on portfolio flow data to measure cross-border flow co-movement, and show NBFI flows are more connected.

Channel 3: Global Financial Cycle

We extract the global factor (Miranda-Agrippino & Rey 2020) from a panel of risky asset returns using PCA. Panel regressions test whether NBFI penetration amplifies GFC transmission to local credit/asset prices. IV estimation instruments the GFC with US monetary policy shocks (Bruno & Shin 2015).

$$y_{\{c,t\}} = \alpha_c + \beta_1 * GFC_t + \beta_2 * NBFI_{\{c,t\}} + \beta_3 * (GFC_t \times NBFI_{\{c,t\}}) + \gamma * X_{\{c,t\}} + \epsilon_{\{c,t\}}$$

If β_3 is significant, NBFI penetration amplifies the global financial cycle.

Cross-Country Quantile Connectedness [Key Innovation]

We estimate a quantile VAR across country equity returns + USD funding at $\tau = \{0.05, 0.50, 0.95\}$. Left-tail connectedness is expected to be substantially higher than median connectedness --contagion is asymmetric and activates during stress.

Contagion Cascade Simulation

We simulate a specific scenario: a major US hedge fund collapse. The cascade propagates through prime brokerage losses, reduced credit to other HFs, cross-border deleveraging, EME capital outflows, and currency depreciation --creating a multi-round feedback loop.

4.4 Expected Contributions

1. A complete model of the cross-border NBFI contagion network --going beyond the US-to-EME focus of Banerjee et al. (2025) to map the full bilateral structure.
2. Evidence on asymmetric contagion: quantile connectedness reveals that cross-border spillovers are much stronger during stress ($\tau=0.05$) than in normal times.
3. Location-scale evidence that NBFI-related capital flows increase the conditional variance of EME financial variables --creating fatter tails.
4. A scenario-based cascade model calibrated to the Archegos/LTCM template, showing how prime brokerage concentration creates cross-border propagation channels.

5. Shared Econometric Toolkit

All three projects draw on a common econometric toolkit implemented in Python. The toolkit provides two main modules:

5.1 Quantile VAR Module (`src/econometrics/quantile_var.py`)

- Quantile VAR estimation at arbitrary tau (Cecchetti & Li 2008, Montes-Rojas 2019)
- Quantile impulse response functions (QIRFs) via companion-form representation
- Quantile connectedness: Diebold-Yilmaz at tail quantiles (Ando et al. 2022)
- Rolling quantile connectedness for tracking tail-risk evolution over time
- Growth-at-Risk regressions (Adrian, Boyarchenko & Giannone 2019)
- GaR term structure across horizons and quantiles

5.2 Location-Scale Module (`src/econometrics/location_scale.py`)

- Two-step location-scale: OLS for $E[y|X]$, then OLS for $\log \text{Var}[y|X]$
- Conditional quantile construction from fitted location and scale
- Skewed-t location-scale density by MLE (Hansen 1994 / Adrian et al. 2022)
- Tail risk amplification tests: interaction effects across the quantile spectrum
- Variance ratio tests (Levene, Brown-Forsythe) for heterogeneous volatility
- Location-scale Growth-at-Risk with NBFI amplification channel

5.3 Existing Modules (from Phase 1)

- Network analysis: centrality, contagion matrices, bipartite projections (`src/analysis/network.py`)
- Systemic risk: CoVaR, MES, SRISK, Diebold-Yilmaz connectedness (`src/analysis/systemic_risk.py`)
- DCC-GARCH: dynamic correlations, sector averages, breakdown tests (`src/models/dcc_garch.py`)
- VaR-constraint: procyclicality, fire-sale simulation (`src/models/var_amplification.py`)
- NBFI subsectors: LDI spirals, MMF runs, prime brokerage contagion (`src/models/nbfi_subsectors.py`)
- Global financial cycle: PCA, panel regressions, IV estimation (`src/analysis/global_financial_cycle.py`)

6. Implementation Status & Codebase

Repository Structure

```
SergioSola/  
  src/  
    analysis/          # Network, systemic risk, GFC modules  
    models/            # DCC-GARCH, VaR-constraint, NBFI subsectors  
    econometrics/      # [NEW] Quantile VAR, location-scale  
    data/              # Data loaders and builders  
    utils/             # Configuration  
  notebooks/  
    project1_shadow_leverage/ # Project 1 notebook  
    project2_mp_transmission/  # Project 2 notebook  
    project3_cross_border/     # Project 3 notebook  
  reports/             # PDF reports  
  scripts/             # Report generation scripts  
  tests/               # Unit tests
```

Module Summary

Module	Lines	Key Functions	Used By
quantile_var.py	~380	estimate_quantile_var, quantile_irf, quantile_connectedness, growth_at_risk	P1, P2, P3
location_scale.py	~380	location_scale_model, tail_risk_amplification, skewt_fit, va	P1, P2, P3
systemic_risk.py	~470	estimate_covar, compute_connectedness, rolling_connectedness, granger_network	P1, P2, P3
network.py	~210	build_exposure_network, centrality, bipartite_projection, cc	P1, P3
dcc_garch.py	~345	fit_all_garch, estimate_dcc, sector_correlations, breakdown_test	P2
var_amplification.py	~455	procyclicality, fire_sale_simulation, run_counterfactual, am	P2
global_financial_cycle.py	~370	extract_global_factor, panel_gfc_reg, iv_gfc_regression, connectedness_amp	P3
nbfi_subsectors.py	~550	LDI_spiral, MMF_run, PB_contagion, build_hedge_fund_se	P3

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